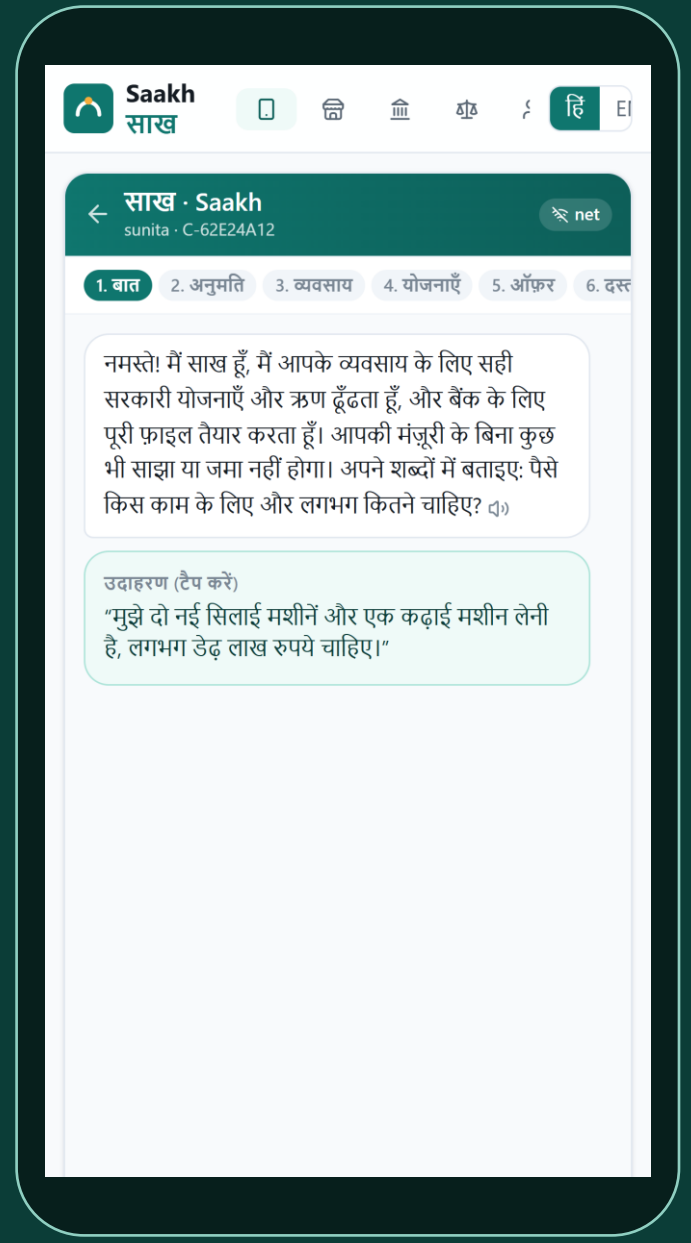
TECH4BHARAT 2026 · PROBLEM 3.2 FORMAL CREDIT AND SCHEMES · AGENTIC AI

Saakh साख

From UPI trail to bank-ready credit

An AI agent that helps India's invisible micro enterprises reach formal credit and the government schemes built for them, with their consent, a cited source for every claim, and their approval before anything is sent.



Team Soliloquy
Sumit Mukherjee (Team Lead), IIT Kharagpur

Viable businesses, invisible to banks

Meet Sunita, a tailor in Barabanki

- Stitches for about 60 paying customers a month, mostly paid by UPI
- No ITR, so the bank cannot see her
- Qualifies for PM Vishwakarma and MUDRA; nobody has told her
- Repays a moneylender ₹1,200 every Monday on a ₹50,000 loan, about 57% a year

The barrier is not creditworthiness. It is paperwork, scheme complexity and months of branch visits.

₹30 lakh cr

addressable MSME credit gap, about 24% of demand

SIDBI with Crisil, May 2025

7.6 crore

MSMEs on the Udyam portal, 99.3% of them micro

Parliamentary Standing Committee, Mar 2026

24 to 36%

interest paid to moneylenders when formal credit fails

Tech4Bharat problem brief

40 to 50%

of PMEGP applications rejected by banks, driven by credit-score reliance

Parliamentary Standing Committee, Mar 2026

One agent, about two minutes, a bank-ready file

1

Talk

Hindi, Hinglish or English, by voice

2

Consent

4 separate, purpose-bound artefacts

3

Understand

Cash flow, seasons, moneylender loans

4

Match

10 scheme pathways, every rule cited

5

Ask

Only questions that change an outcome

6

Offer

A loan sized to lean and busy months

7

Build

17 papers become 9 items, DPR generated

8

Approve

The owner approves the exact file

9

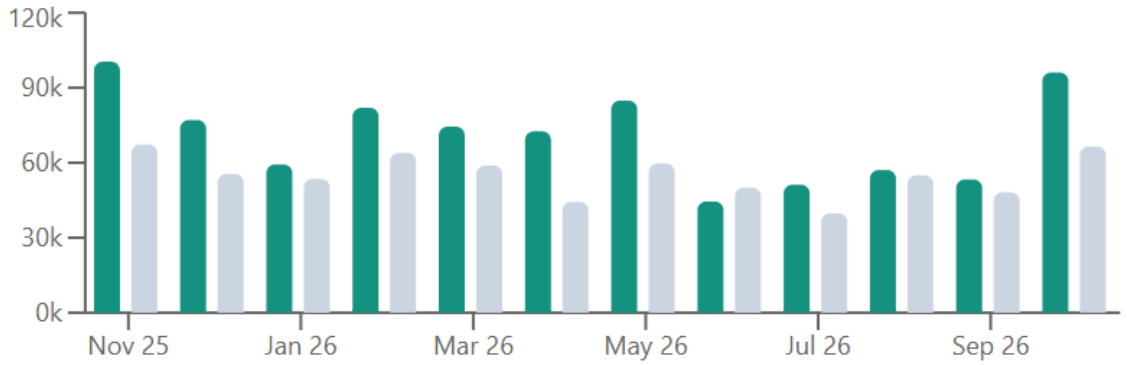
Track

The branch gets a pre-appraised file

Built on India's digital rails: UPI, Account Aggregator, DigiLocker, Udyam. In the prototype these are simulated adapters with production-shaped interfaces, labelled SIMULATED.

It reads the business, not the paperwork

From your account (AA) 12 months SIMULATED



Average business income ₹71,043/month. 93% received digitally from 60 paying customers a month.

Seasonal business. Peak: Nov, Oct, May. Lean: Jun, Jul, Sep (lean month ≈ 60% of a normal month).

Likely moneylender repayments. 52 weekly payments of ₹1,200 to RAMKUMAR YADAV (≈ ₹62,400/year). Refinancing this is part of the plan.

Moneylender found

52 weekly payments of ₹1,200 to one person: flagged as an informal loan, about 57% a year, and refinanced in the offer.

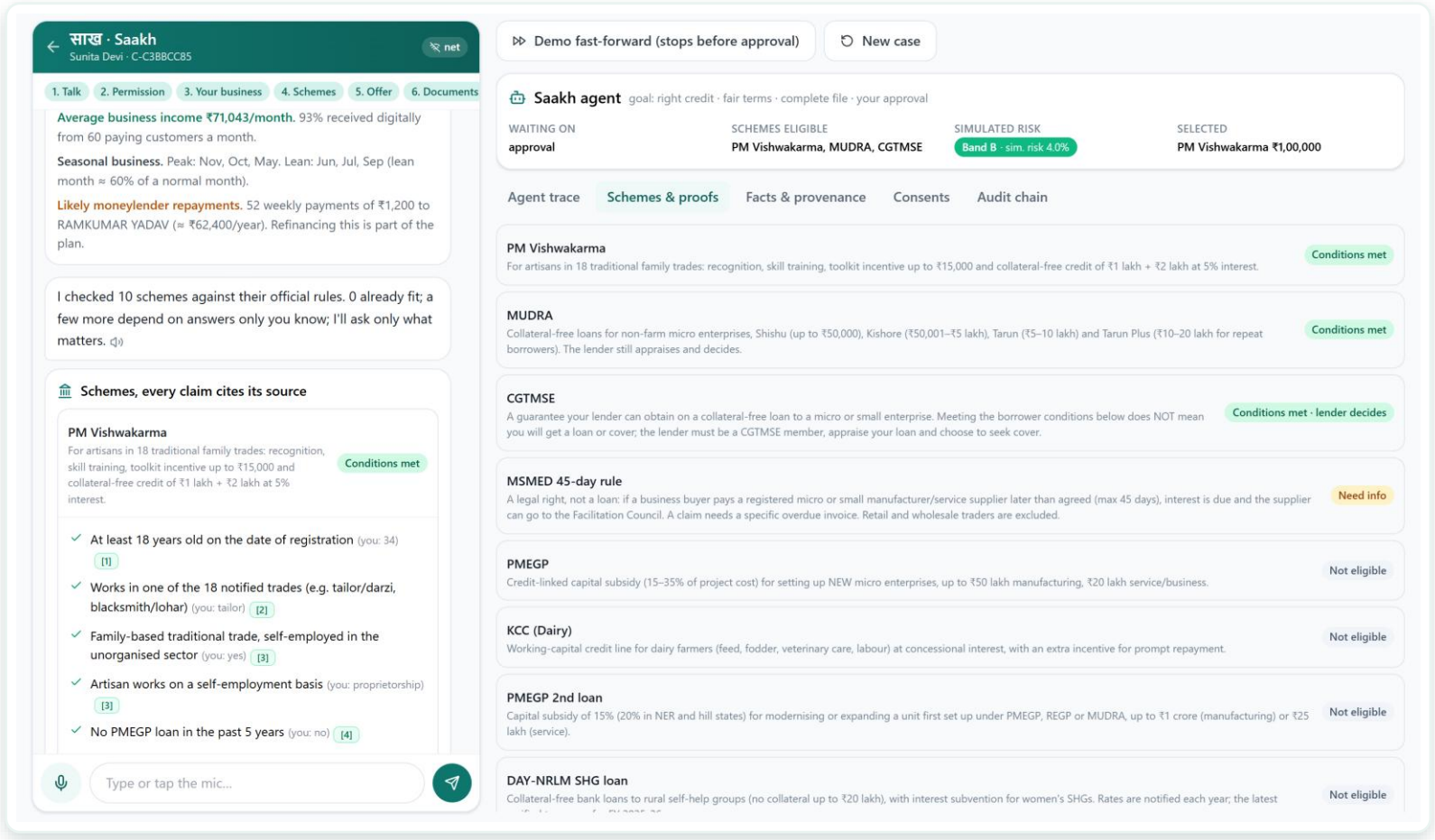
Seasons, not averages

Lean months run at about 60% of normal, so the repayment plan shrinks when income does.

Evidence, minimal data

Turnover, B2B buyers and bounces are derived with evidence. Account Aggregator data is processed in memory and never stored.

Every scheme claim comes with its source



Conditions met

every criterion passes, each with its citation

Provisional

rests on a self-declaration an agency verifies

Needs info

the agent asks the one most decisive question

Lender decides

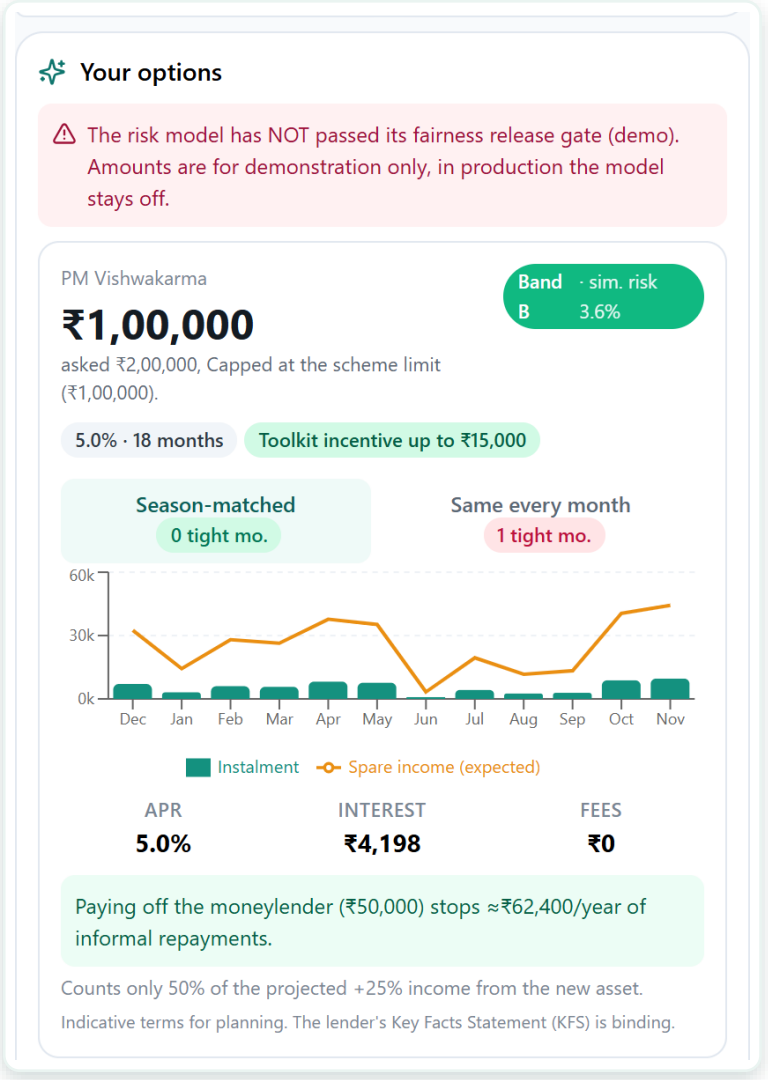
e.g. CGTMSE cover is the bank's choice

Not available

e.g. Stand-Up India, period ended March 2025

10 scheme pathways · rules are data (YAML) · the loader refuses any rule without a citation · unknowns never become "eligible"

A loan sized to the seasons, not a fixed EMI

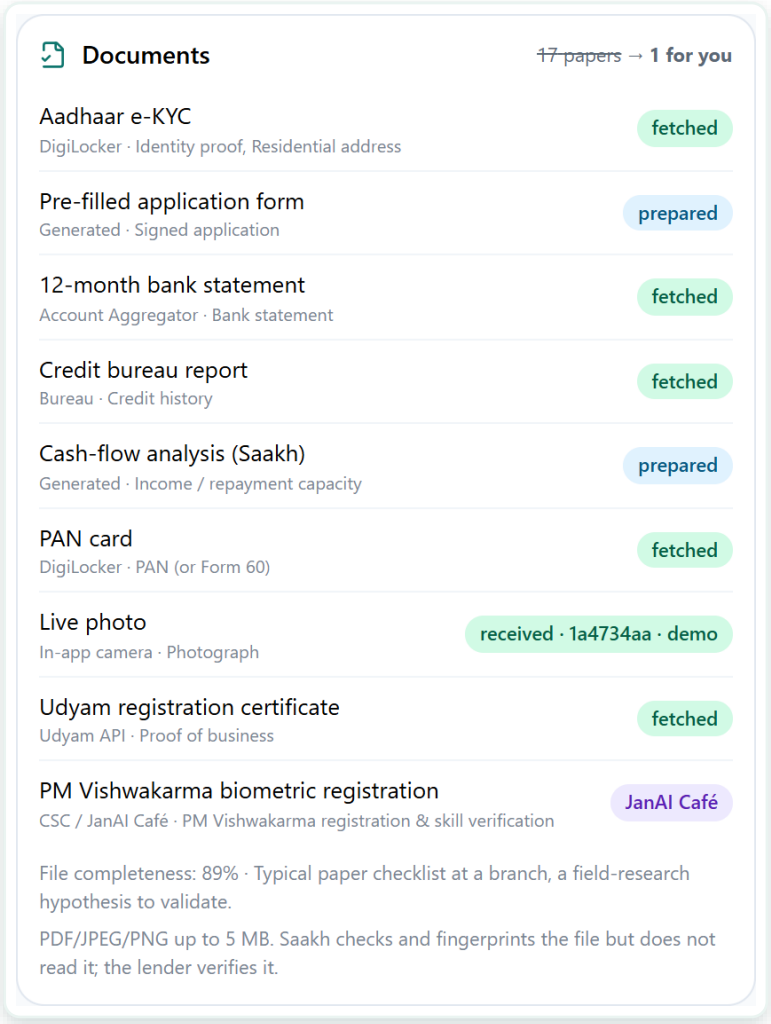


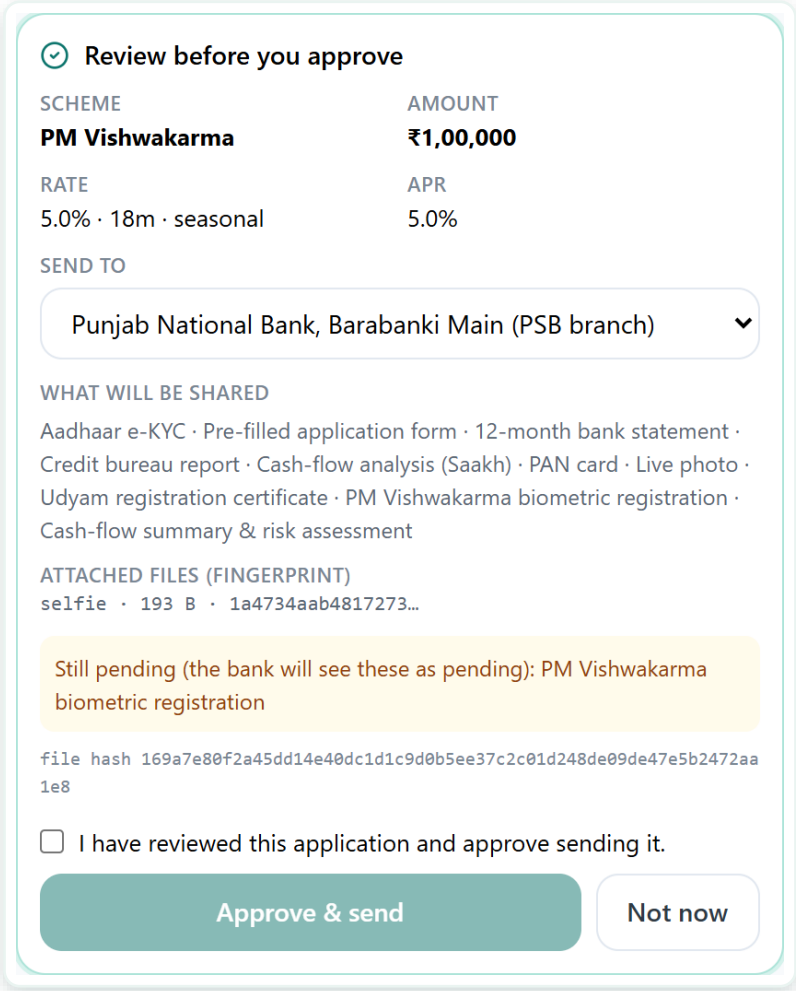
12 → 0

tight months for a dairy unit over a 36-month loan: same-every-month EMI vs a season-matched schedule with the same present value

- Sized from disposable cash flow and risk policy, with only half of the projected uplift from the new asset
- Key Facts Statement-style APR, interest and fees, shown before choosing
- Refinances the moneylender: for Sunita, it stops about ₹62,400 a year of informal repayments
- An honest "not now" with a path back, when a loan would strain the business

From 17 papers to 9 items, approved as an exact file





Minimum-effort set cover

Most items are fetched or generated: pre-filled form, cash-flow analysis, project report PDF. Physical steps go to a JanAI Café.

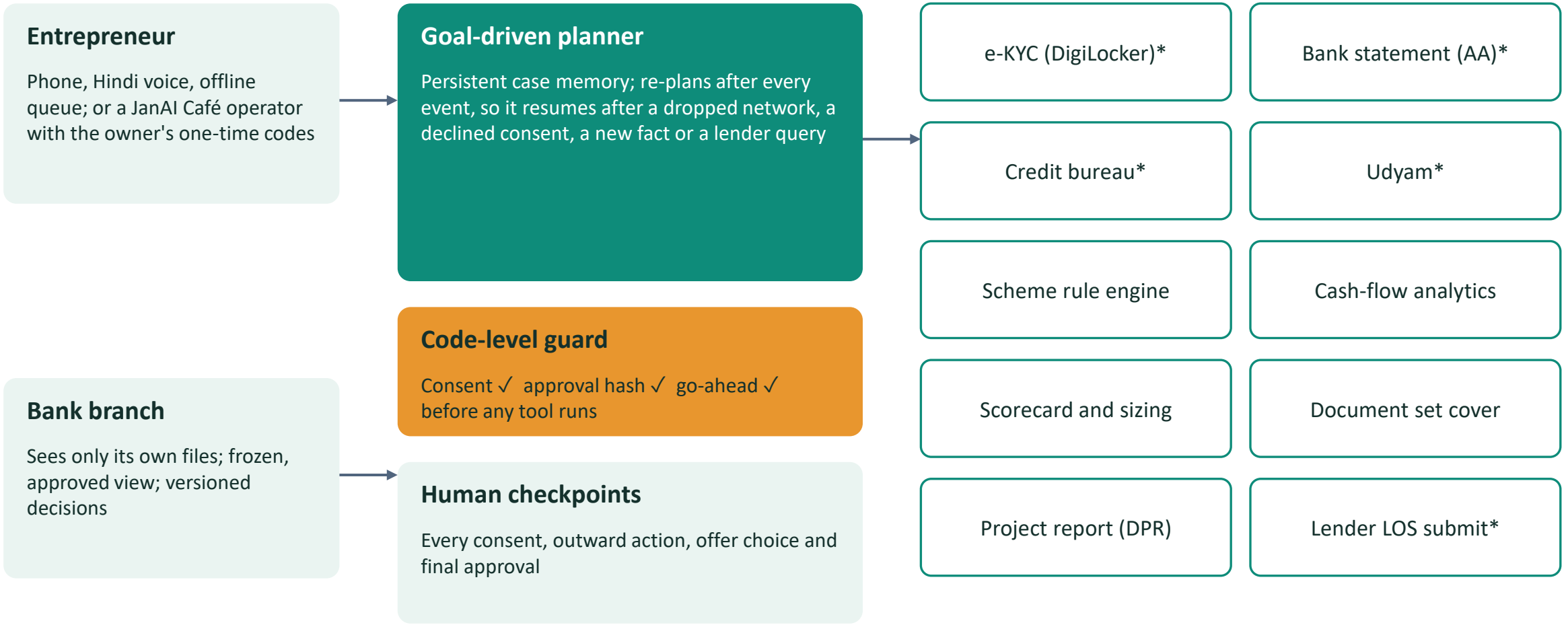
Bound to a fingerprint

Approval covers the SHA-256 of the file and of every attachment. Any change voids it; a sent copy is superseded until re-approved.

Revocable

Withdrawing consent withdraws the file and erases its contents from the bank's copy.

A real agent: plans, acts, recovers, and asks the human



*Recovery: the bank-data call times out by design; the agent retries with backoff, then falls back to a statement upload or a café visit. * simulated adapters. Eligibility, risk and amounts come from auditable engines, never from generated text.*

Trust is enforced in code, not promised

Purpose-bound consent

ReBIT-style artefacts with expiry. Decline any one and the agent asks instead.

Revocation that deletes

Withdrawing or expiry erases derived data, chat cards and the bank's copy, on a schedule.

Tamper-evident audit

A hash-chained log that stores no personal or financial values.

Exact-file approval

SHA-256 binds the file and attachments; café codes are one-time, per step, 10-minute.

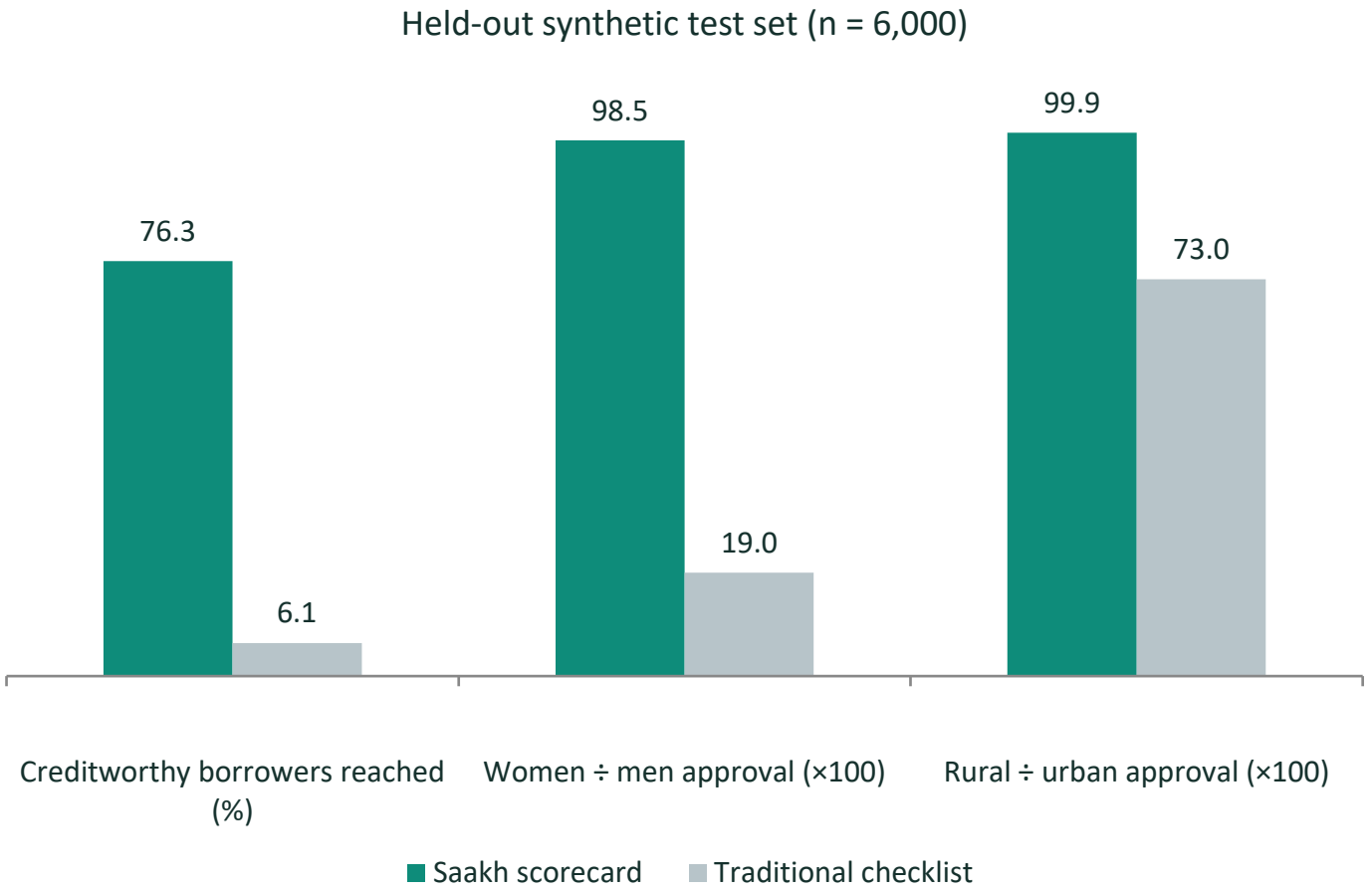
Lender isolation

Each branch sees only its files; decisions are versioned, final and idempotent.

Strict inputs

Every event is type-checked; actions are accepted only at their checkpoint.

Fairer than the checklist, and honest when a gate fails



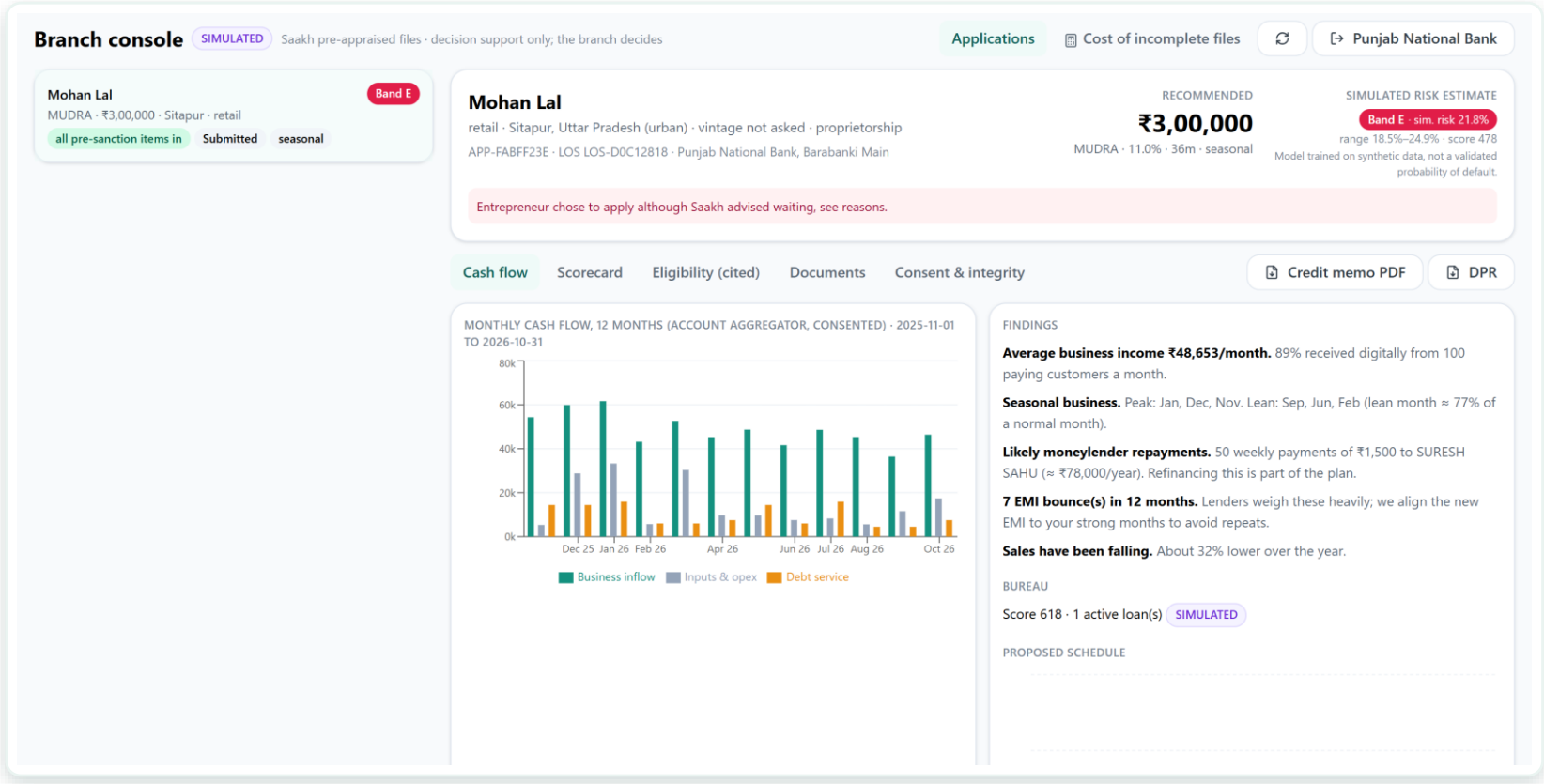
BLOCKED, by design

1 of 7 fairness gates fails (ST vs General, −7.3 pp on the test split; −1.85 pp on a fresh draw). The rule fails closed: production refuses score-based offers until the gap is explained.

- Monotone, explainable points table; reason codes for every score
- Gender, caste, religion and location are never inputs
- AUC 0.736, calibration error 0.6%; a boosted-tree challenger gains nothing

Labels come from a synthetic generator: these numbers show the pipeline works, not real-world performance. Lender data is needed before any real use.

For the bank: a pre-appraised file it can trust



≈ ₹79,000

rework cost avoided per branch per month (about 105 officer-hours)

- Credit memo and project report PDFs
- Reason codes, cited eligibility, integrity proofs
- Fingerprint-checked document downloads
- Declines need the bank's own reasons

Rework estimate uses editable assumptions (60 files a month, 45% incomplete today vs 5%, 2.2 rework cycles of 1.6 hours, ₹750 per officer-hour) to be replaced with field measurements.

Free for the owner, paid for by those who save



Entrepreneur

Pays nothing

Removes the 10 to 15% agent cut and months of visits.



Banks and NBFCs

₹150 to 400 per complete file + success fee

Rework alone costs about ₹1,300 per file today.



DICs and scheme agencies

Programme and outreach budgets

Cohort screening helps them meet PMEGP and PM Vishwakarma targets.



SHG federations, BCs, JanAI Cafés

Assisted-service fee per journey

Screen 40 members in seconds; earn per completed file.

Near-zero marginal cost per file, so growth is bounded by distribution, not compute.

Large market, national rails, local jobs

₹30 lakh cr

MSME credit gap

7.6 cr

MSMEs on Udyam

99.3%

of them micro

Go to market

1. Pilot: one bank branch and one SHG federation
2. Regional rollout with that lender
3. National: AA, ULI, DigiLocker rails and the CSC / JanAI Café network

Scales without rewrites: a new scheme is a YAML file with citations; the scorecard refits per lender; more Indian languages plug into the same parser design.

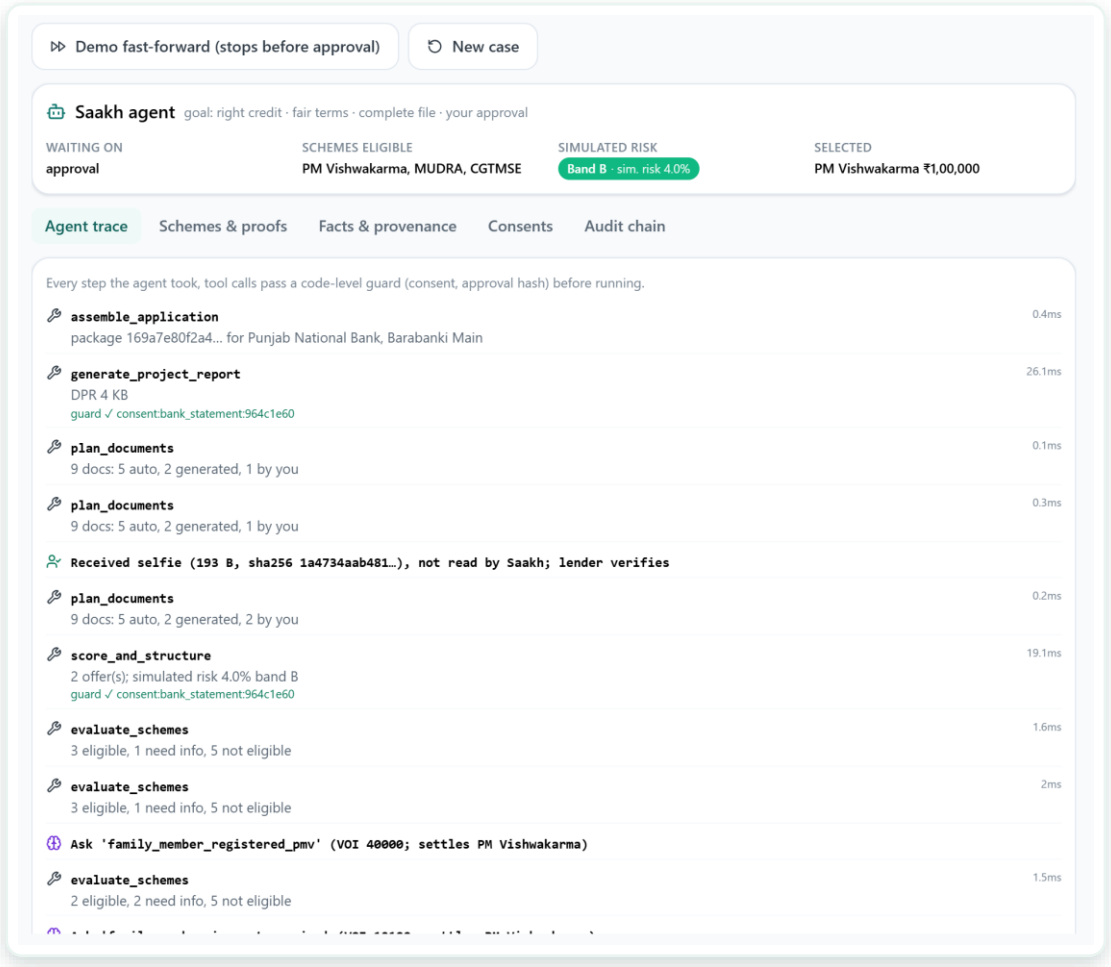
Jobs

Direct: café and business-correspondent operators paid per assisted journey; field verification for physical steps

Indirect, the real lever: every eligible-but-unreached unit that gets funded can hire. PMEGP alone reports about 87 lakh jobs from 10.7 lakh units since 2008-09

Source: Parliamentary Standing Committee on Industry, Mar 2026

Working today, end to end



126

automated tests, including concurrency and consent-lifecycle cases

10

scheme pathways with cited official sources and verification levels

6

synthetic personas, Hindi and English, voice in and read-aloud out

6

rounds of independent code review; every finding and verdict published

Also in the prototype: offline queue, JanAI Café assisted mode, branch console, governance view, SHG federation screening. Stack: FastAPI, React PWA, SQLite, numpy/scipy scorecard. Integrations without open APIs are simulated and labelled.

What we will measure next



Field research plan

- 15 to 20 owners across six business types, 3 to 5 branch managers, 2 to 3 loan agents, one SHG federation or café
- The organisers' six questions become hypotheses, each tied to a product decision it would change
- Branch time-and-rework log per file, to replace our cost assumptions with measured numbers
- Funnel and consent-decline rates measured by the product itself

Next 30 days

Interviews and branch time study; refine questions and document weights

By the finale

Pilot design with a branch and an SHG federation; measured rework numbers

2027

AA and ULI sandbox integration; scorecard refit on lender data with the same fairness gates; more languages

Saakh turns trust already earned into formal credit



Agentic

Plans, uses guarded tools, recovers from failure, and stops for the human at every consequential step



Trusted

Consent, cited proof, exact-file approval and revocation, all enforced in code and tested



Viable

Free for owners, paid by lenders who save on rework; scales on India's digital rails

github.com/sumitmukherjee7/saakh

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